

Jenil Shingala

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Education

Bachelor of Science in Computer Science

Jan 2022 - Dec 2025

California State University, Sacramento | Major GPA 3.8 | Dean's Honor List

- **Coursework:** Parallel & Concurrent Programming, Machine Learning, Artificial Intelligence, Operating Systems, Computer Architecture, Computer Networks, Data Structures & Algorithms
- **Certifications:** NVIDIA-Certified Associate: AI Infrastructure & Operations (NCA-AIIO) (NCCL, BlueField DPU, TensorRT, RoCE/InfiniBand, MIG, vGPU) [LINK](#); NVIDIA-Certified Associate: Generative AI LLMs (NCA-GENL) [LINK](#)

Technical Skills

Languages	Python, C, C++ (STL, GDB, Valgrind), JavaScript, CUDA
ML / GPU Systems	PyTorch, TensorFlow, Triton, GPU Kernel, NCCL, DDP, HPC, Multi-GPU, RoCE/InfiniBand
AI / LLM	LLMs, Agentic RAG, GenAI, OpenCV, RNN, CNN, NLP, Transfer Learning, Model Evaluation
Platform & Tools	AWS, Azure, Docker, Jenkins, CI/CD, Git/GitHub, MySQL, Postman, Grafana, Linux, Agile

Experience

AI Software Engineer Intern (Project Lead)

May 2025 - Dec 2025

Advanced Integration & Controls | Sacramento, CA

- Architected and shipped a production **agentic RAG system** that grounds LLM responses in verified documents, lifting answer accuracy **96%** over the team's prior ungrounded baseline.
- Engineered the retrieval layer on **Azure AI Search** with **vector embeddings** and **Blob Storage** for low-latency semantic search, driving **87%** higher daily productivity across a 10-25 engineer team.
- Automated cross-team document workflows (Bill of Materials, PCN) via an **MCP integration** to Monday.com and Power Automate, unifying **5+** previously manual data sources.

ML/Software Engineer

Sep 2024 - May 2025

SentrySight LLC | Sacramento, CA

- Deployed a full-stack **React/AWS** (RDS, LightSail) computer-vision app serving real-time inference at **<50ms latency**.
- Designed a two-model deep-learning pipeline (**YOLOv11n** + transfer-learned **CNN**) that reached **91% CNN accuracy** and **94% face-recognition accuracy** across a **500+ person database**.
- Owned the system end-to-end from design through **CI/CD** deployment, building a **15+** check QA framework with human-in-the-loop review to catch quality drift before release.

Embedded Engineer Intern

Aug 2021 - Dec 2021

Goldfield Weighting Solution

- Designed and shipped **C++** embedded firmware for industrial weighing systems, applying **object-oriented design patterns** at the hardware-software boundary to support production deployment.
- Partnered with a cross-functional team of **4** on **hardware integration**, connecting firmware to physical scale hardware to enable reliable end-to-end operation.
- Diagnosed and resolved hardware-software defects through systematic end-to-end testing, cutting the **defect rate by 23%** ahead of release.

Projects

FlashAttention GPU Kernel (from scratch) | Triton · PyTorch · CUDA · GPU Kernels · Online-Softmax. [GITHUB](#)

- Implemented **FlashAttention** as a fused **Triton** GPU kernel using **online-softmax** streaming recurrence, cutting peak attention memory from **$O(N^2)$** to **$O(N)$** **250x lower** at sequence length 8192.
- Benchmarked against PyTorch's native **scaled_dot_product_attention** using **CUDA-event timing** and peak-memory profiling, confirming **~17MB vs ~4.3GB** peak memory; extended with **Triton autotuning** and causal masking.

AI Hackathon 2025 @ Hornet Hacks: Ticker Skimmer | Python · Bidirectional LSTM · RNN · NLP · Gradio. [GITHUB](#)

- Built a stock predictor fusing Reddit sentiment (NLP) with a **bidirectional LSTM** time-series model and a **JIT optimization engine**, served through a real-time **Gradio** UI.
- Engineered **5-day time-step sequences** with **Z-score normalization** and **EarlyStopping** to capture pattern reversals from both past and future context.

Network Intrusion Detection System | Python · FCNN · CNN · Keras Tuner · TensorFlow. [GITHUB](#)

- Trained a deep-learning intrusion-detection model (**FCNN/CNN** in **TensorFlow**) to **97% accuracy** on imbalanced network-traffic data.
- Tuned hyperparameters with **Keras Tuner** and validated using **precision (BLEU)**, **recall (ROGUE)**, **F1**, and confusion-matrix analysis to handle class imbalance.

AI Hackathon 2024 @ Hornet Hacks: Weapon, Face Detection | YOLOv11n · OpenCV · CNN · Inception v3 · CUDA. [GITHUB](#)

- Built a two-model **computer-vision pipeline** (YOLOv11n detection + **CNN** via **transfer learning** from Inception v3) reaching **94% detection accuracy**.
- Cut false positives **87%** via dual-model agreement at **<50ms latency** across a **500+ database (91% CNN accuracy)**.